A DATA-DRIVEN ANALYSIS MODEL FOR KENYA'S HEALTH FACILITY LANDSCAPE USING SOCIAL HEALTH AUTHORITY DATA.

DECLARATION

I hereby declare that this Project Proposal is my own work and has, to the best of my knowledge, not been submitted to any other institution of higher learning for any award.

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APPROVAL

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ABSTRACT

Achieving Universal Health Coverage in Kenya requires strategic planning based on accurate spatial analysis of healthcare infrastructure. This project proposes the development of a data-driven analytical model to assess Kenya's health facility landscape using official data from the Social Health Authority (SHA). The model will transform raw facility data into actionable intelligence through a systematic approach involving data ingestion, spatial analysis, and interactive visualization. The methodology will employ Python programming with libraries including Pandas for data processing, GeoPandas for spatial analysis, and Streamlit for dashboard development. Key components include a data preprocessing engine for cleaning SHA datasets, a spatial equity module for calculating facility-to-population ratios, and a public-private analysis module for evaluating healthcare provision patterns. Expected outcomes include an interactive dashboard that visualizes facility distribution, identifies underserved regions, and provides evidence-based recommendations for resource allocation. The project will follow a structured development lifecycle, from requirements analysis to implementation and testing. It is anticipated that this model will serve as a valuable decision-support tool for healthcare planners and

policymakers, contributing to more equitable healthcare resource distribution across Kenya's counties. The successful implementation of this project will demonstrate the practical application of data science in addressing real-world public health challenges while providing a comprehensive learning experience in system development and spatial analytics.

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1. CHAPTER 1 – INTRODUCTION

1.1. Background to the Study

The achievement of Universal Health Coverage (UHC) stands as a primary objective within Kenya's healthcare policy framework, aiming to ensure equitable access to quality essential health services without financial hardship for all citizens. A fundamental prerequisite for attaining this goal is the strategic and equitable distribution of health facilities across the nation. Kenya's health system is characterized by a pluralistic model comprising facilities owned and operated by diverse entities including the Ministry of Health, county governments, private for-profit organizations, faith-based institutions, and non-governmental organizations. This heterogeneous landscape presents both opportunities and challenges for coordinated health service delivery.

A critical challenge in health sector planning is the transformation of raw administrative data into actionable intelligence for evidence-based decision-making. The Social Health Authority (SHA), established under Kenya's health reforms, serves as the national custodian of critical health system data, including a comprehensive registry of health facilities. This dataset represents a significant national asset with immense potential to inform policy. However, in its raw, tabular form, the data offers limited utility for understanding complex spatial patterns, identifying coverage gaps, or simulating the impact of strategic interventions.

The field of data science offers powerful methodologies to address this gap. Techniques in geospatial analysis, statistical modeling, and interactive data

visualization enable the creation of sophisticated analytical models. Such a model can systematically evaluate the density, distribution, equity, and service capacity of health facilities relative to population needs. Therefore, this project is situated at the intersection of public health informatics and data science, proposing to leverage the SHA dataset to build a decision-support tool that can enhance the planning and management of Kenya's health facility infrastructure.

1.2. Statement of the Problem

Current approaches to analyzing Kenya's health facility landscape often rely on descriptive statistics and static reports generated from isolated datasets. These methods are insufficient for proactive, strategic planning as they fail to provide a holistic, integrated, and interactive view of the healthcare infrastructure. Consequently, policymakers, county health executives, and planners lack a unified tool to answer critical, data-driven questions such as: Which counties exhibit the most severe disparities in healthcare access based on facility-to-population ratios? How does the mix of public and private healthcare provision vary geographically, and what are the implications for service affordability and UHC? Where should new infrastructure investments be prioritized to maximize population coverage and reduce existing inequities?

In the absence of a dedicated analytical model, the rich, detailed data housed by the SHA remains underutilized. This data-underuse leads to suboptimal resource allocation, persistent geographical and socioeconomic inequities in healthcare access, and an inability to perform predictive scenario analysis for infrastructure development. There exists a significant gap between data availability and strategic insight. This project, therefore, seeks to address this gap by designing and developing a robust, data-driven analytical model that synthesizes SHA data with other relevant datasets to generate quantifiable metrics, visual insights, and evidence-based recommendations for improving Kenya's health facility landscape.

1.3. Proposed Solution

This study proposes the design and development of a modular, data-driven analytical model named the Kenya Health Facility Landscape Analysis Model (KHF-LAM). The model will utilize the official health facility dataset from the Social Health Authority as its primary input. The solution is conceptualized as a software-based analytical framework comprising several integrated modules: a Data Ingestion and Harmonization Engine to preprocess raw data; a Spatial Equity and Access Scoring Module to compute and map disparities in healthcare coverage; a Public-Private Provision Analysis Module to evaluate the composition of service delivery; and a Resource Optimization Simulator to propose data-informed recommendations for new facility placement or upgrades. The final deliverable will be an interactive web-based dashboard built using the Streamlit framework, which will visualize all findings and provide an intuitive interface for stakeholders to explore data, filter views, and conduct basic scenario analyses.

1.4. Objectives

1.4.1 General Objective

To design and develop a data-driven analytical model that evaluates the distribution, equity, and service capacity of health facilities in Kenya using data from the Social Health Authority, thereby supporting evidence-based strategic planning in the health sector.

1.4.2 Specific Objectives

1. To design and implement a data preprocessing engine for ingesting, cleaning, validating, and harmonizing the raw SHA health facility dataset.
2. To develop a spatial equity analysis module that calculates facility-to-population ratios, generates accessibility scores, and creates visualizations of distribution patterns across all 47 counties.
3. To create a public-private partnership (PPP) analysis module that evaluates the market structure of health service provision and

identifies regions with high potential for targeted investment or intervention.

4. To design and deploy an interactive dashboard that integrates the model's outputs, providing a user-friendly tool for data exploration, visualization, and the generation of actionable reports.

1.5. Significance of the Study

The study holds significance for multiple stakeholders in Kenya's health ecosystem. For national and county government policymakers, including officials at the SHA and Ministry of Health, the model will provide an evidence-based tool for identifying underserved regions, optimizing resource allocation, and planning new health infrastructure, directly supporting UHC goals. County health departments will benefit from a tailored view of their jurisdiction's facility landscape, enabling better local planning and advocacy. Researchers and academicians in public health and data science will gain access to a novel methodological framework and an open-source analytical tool that can be extended or adapted for further studies. Development partners and NGOs operating in the health sector can use the insights to target their programs more effectively. For the student researcher, this project represents a capstone application of data science, software engineering, and systems analysis skills to a complex, real-world problem, significantly enhancing technical proficiency and problem-solving capabilities.

1.6. Scope of the Study

The scope of this study is explicitly defined by the following boundaries. The data scope is limited to the health facility datasets available on the SHA website (https://sha.go.ke/resources/categories/4) as of the 2024 academic year and publicly available county population data from the Kenya National Bureau of Statistics (KNBS). The functional scope encompasses the development of the KHF-LAM analytical model and its accompanying interactive dashboard. The model will focus on analyzing facility attributes (type, ownership, location), their

spatial distribution, and derived metrics like population coverage. It will not include analysis of patient-level clinical data, real-time facility utilization or occupancy rates, staff-to-patient ratios, detailed financial costing of recommendations, or the development of a real-time, transactional system. The dashboard will be a prototype for demonstration and analytical purposes.

1.7. Assumptions

The study is premised on several key assumptions:

1. The data provided by the SHA is substantially accurate, current, and representative of the operational health facility landscape in Kenya.
2. The geographical coordinates (latitude and longitude) for facilities within the dataset are correct and can be reliably used for spatial analysis.
3. The population data obtained from KNBS is accurate and appropriate for calculating county-level facility-to-population ratios.
4. A facility's listing in the SHA registry is synonymous with it being operational and providing services.
5. The availability of standard software tools and libraries (Python, relevant packages) and adequate computational resources for development and testing.

1.8. Limitations and Delimitations

The project acknowledges certain limitations. The accuracy and completeness of the model's outputs are contingent upon the quality of the input SHA data, which may contain inconsistencies, missing values, or outdated information. The spatial accessibility metrics may be simplistic, as they do not account for real-world travel conditions, terrain, or transportation networks due to data constraints. To mitigate these limitations, the study will employ rigorous data validation and cleaning procedures, clearly document all data assumptions, and use established, transparent methodologies for calculating metrics.

The study is deliberately delimited in the following ways: it is a static analysis based on a snapshot of data at a point in time rather than a dynamic, real-time monitoring system; it focuses on facility presence and distribution rather than the quality of services rendered; and it is designed as a descriptive and diagnostic tool to inform planning, not an operational tool for daily facility management.

1.9. Definition of Terms

1. Data-Driven Model: A computational and statistical framework that uses data as the fundamental input to identify patterns, perform analysis, and generate predictive or prescriptive insights.
2. Health Facility Landscape: The comprehensive configuration of health facilities within a defined geographical area, encompassing their number, types, ownership, spatial distribution, and service capabilities.
3. Spatial Equity: The degree to which health resources (e.g., facilities) are distributed fairly across a geographical area in proportion to the needs of the population residing in different parts of that area.
4. Social Health Authority (SHA): A state corporation in Kenya established under the Health Act, responsible for managing the national health insurance fund and maintaining key health sector databases.
5. Interactive Dashboard: A data visualization tool that displays key performance indicators (KPIs) and metrics, allowing users to interact with the data through filters, drill-downs, and other controls to explore information dynamically.
6. Universal Health Coverage (UHC): A World Health Organization (WHO) goal that ensures all people have access to the full range of quality health services they need, when and where they need them, without financial hardship.

2. CHAPTER 2 – LITERATURE REVIEW

2.1. Introduction

This chapter presents a comprehensive review of existing literature, systems, and methodologies related to health facility mapping, spatial analysis in healthcare, and data-driven decision support systems. The review explores both global and local contexts to establish the theoretical foundation for developing a data-driven analytical model for Kenya's health facility landscape. It examines the evolution of Geographic Information Systems (GIS) in healthcare planning, existing health facility assessment tools, and identifies gaps that the proposed model aims to address.

System Name	Developer/Country	Key Features	Limitations	Relevance to Proposed Solution
Kenya Master Health Facility List (KMHFL)	Ministry of Health, Kenya	• National registry of all health facilities • Facility categorization by type and level • Basic search and filtering capabilities	• Static data presentation • No spatial analysis tools • Limited data visualization • Manual updates required	Provides baseline data for analysis; Proposed model will add analytical capabilities to this foundational data
WHO Health Facility Registry	World Health Organization (Global)	• International standardization • Cross-country comparability • Integration with global health indicators	• Generic framework not customized for Kenya • Limited local context integration • Slow update cycles	Offers international best practices that can be adapted for the Kenyan context in the proposed model
AfyaMap Tanzania	Ministry of Health, Tanzania	• Interactive web-based mapping • Service coverage analysis	• Country-specific to Tanzania • Limited predictive analytics	Demonstrates successful implementation of GIS in East Africa; Provides design insights

		• Integration with population data	• Requires significant customization for Kenya	for Kenyan adaptation

2.2. Related Systems (2-3)

2.2.1. World Health Organization (WHO) Health Facility Registry

The WHO maintains a global health facility registry that serves as a standardized directory of health facilities worldwide. This system provides basic information about facility locations, types, and services offered. The registry is built on the System of Health Accounts (SHA) framework and aims to support health system strengthening through improved facility data management. However, its application in Kenya has been limited by inconsistent data updates and lack of integration with local planning systems. The system primarily serves as a reporting tool rather than an analytical platform for strategic decision-making.

2.2.2. Kenya Master Health Facility List (KMHFL)

The Kenya Master Health Facility List, maintained by the Ministry of Health, represents the official national registry of all health facilities in Kenya. This system captures comprehensive data including facility ownership, services offered, and geographical coordinates. While the KMHFL serves as the primary reference for facility identification, it lacks advanced analytical capabilities. The system provides static reports and basic search functions but does not offer spatial analysis, equity scoring, or predictive modeling features. Users cannot perform comparative analyses across counties or simulate the impact of new facility placements.

2.2.3. AfyaMap (Tanzania Health Facility Mapping System)

Developed for Tanzania's health sector, AfyaMap is a web-based GIS platform that visualizes health facility distribution and service availability. The system integrates multiple data sources including facility locations, population data, and

epidemiological information. AfyaMap provides interactive mapping capabilities, service coverage analysis, and gap identification features. However, its architecture is specific to Tanzania's administrative structure and health system organization, requiring significant adaptation for application in Kenya. Additionally, the system focuses more on visualization than on predictive analytics or optimization modeling.

2.3. Limitations (Weaknesses of these Systems)

Limited Analytical Capabilities: Existing systems primarily function as data repositories with basic query and reporting features. They lack sophisticated analytical modules for spatial equity analysis, predictive modeling, and optimization algorithms that could support strategic planning.

Static Data Representation: Current systems provide snapshots of facility data without dynamic updating or real-time integration with other data sources. This limits their utility for monitoring changes in healthcare access patterns or evaluating the impact of policy interventions over time.

Poor User Interactivity: Most existing health facility systems offer limited user interaction capabilities. Users cannot customize views, perform scenario analysis, or generate tailored reports based on specific planning questions or regional priorities.

Insufficient Integration with Planning Processes: The systems operate in isolation from actual health planning workflows. There is minimal integration with resource allocation systems, budgeting processes, or performance monitoring frameworks, reducing their practical utility for decision-makers.

Lack of Predictive and Prescriptive Features: Current systems are descriptive rather than predictive or prescriptive. They show what exists but cannot forecast future needs, identify optimal locations for new facilities, or recommend specific interventions to address coverage gaps.

2.4. How Your Proposed Solution Will Handle These Weaknesses

The proposed Kenya Health Facility Landscape Analysis Model (KHF-LAM) addresses these limitations through several innovative approaches:

Advanced Analytical Engine: KHF-LAM incorporates modular analytical components including spatial equity scoring algorithms, facility clustering analysis, and optimization models. These features transform raw facility data into actionable intelligence that supports evidence-based decision-making.

Dynamic Data Processing: The system implements automated data validation, cleaning, and updating mechanisms. It can integrate multiple data sources including SHA data, population statistics, and geographical information to create a comprehensive, up-to-date view of the health facility landscape.

Interactive Dashboard Interface: Using Streamlit framework, KHF-LAM provides an intuitive, web-based dashboard that allows users to interact with data through filters, custom visualizations, and scenario simulation tools. This enhances usability and accessibility for non-technical stakeholders.

Integration-Ready Architecture: The model is designed with APIs and export functions that enable integration with existing health planning systems. This facilitates the practical application of analytical insights in budget allocation, infrastructure planning, and performance monitoring processes.

Predictive and Prescriptive Capabilities: Beyond descriptive analysis, KHF-LAM includes modules for projecting future facility needs based on population growth trends and recommending optimal locations for new facilities using multi-criteria decision analysis techniques.

3. CHAPTER 3 – METHODOLOGY

3.1. Introduction

This chapter outlines the systematic approach that will be employed to design, develop, and implement the proposed data-driven analytical model. The methodology follows a structured software development lifecycle, integrating principles from data science, geospatial analysis, and human-computer interaction. The approach ensures that the project objectives are met through rigorous planning, execution, and validation processes.

3.2. Project Design

3.2.1. Development Methodology

The project will adopt the Agile-Scrum methodology with incremental development phases. This iterative approach allows for regular feedback incorporation, adaptive planning, and continuous improvement throughout the development process. The methodology is particularly suitable for this project due to its emphasis on user-centric design, flexibility in responding to changing requirements, and emphasis on delivering working software in short cycles (sprints).

3.2.2. Justification of Methodology

The Agile-Scrum methodology is justified for several reasons. First, the project involves multiple stakeholder perspectives (technical, policy, and user experience) that require continuous alignment. Second, the development of analytical modules benefits from iterative testing and refinement. Third, the methodology supports risk management through regular review and adaptation. Finally, it ensures that the final product closely matches user needs through continuous stakeholder engagement.

3.2.3. System Architecture

The model will follow a modular, three-tier architecture consisting of:

1. Data Layer: Responsible for data ingestion, storage, and preprocessing using SQLite database and Pandas dataframes.
2. Analytical Layer: Contains the core analytical modules (spatial analysis, equity scoring, optimization algorithms) implemented in Python.
3. Presentation Layer: The interactive web dashboard built with Streamlit framework for data visualization and user interaction.

3.3. Design Procedures

The design process will follow these sequential procedures:

1. Requirements Analysis: Conduct stakeholder needs assessment through literature review and consultation with healthcare planning experts to define functional and non-functional requirements.
2. System Design:

- Create data flow diagrams (DFDs) to illustrate information flow through the system.
- Develop entity-relationship diagrams (ERDs) for data modeling.
- Design user interface mockups using wireframing tools.

Create UML diagrams (use case, sequence, and class diagrams) for system specification

3. Module Development: Implement the system in modular components:

- Data preprocessing module
- Spatial analysis and mapping module
- Equity scoring and gap analysis module
- Dashboard and visualization module

4. Integration and Testing: Integrate individual modules and conduct comprehensive testing including unit testing, integration testing, and user acceptance testing.

3.4. System Requirements

3.4.1. Hardware Requirements

3.4.2. - Processor: Intel Core i5 or equivalent (minimum)

3.4.3. RAM: 8GB (minimum), 16GB (recommended)

3.4.4. Storage: 500GB HDD or 256GB SSD

3.4.5. Display: 1366x768 resolution (minimum)

3.4.6. Software Requirements

- Operating System: Windows 10/11, Linux Ubuntu 20.04+, or macOS 10.15+
- Development Environment: Python 3.9+, Jupyter Notebook
- Key Libraries: Pandas 1.4+, GeoPandas 0.12+, Streamlit 1.22+, Plotly 5.14+, Folium 0.14+
- Database: SQLite 3.37+
- Version Control: Git 2.35+

3.5. Data Collection and Analysis Methods and Tools

3.5.1. Data Sources and Collection Methods

1. Primary Data Source: Health facility data will be collected from the Social Health Authority (SHA) website through direct download of CSV/Excel files from the resources portal.
2. Secondary Data Sources:
   - Population data from Kenya National Bureau of Statistics (KNBS)
   - County boundary shapefiles from Kenya Open Data Portal
   - Road network data from OpenStreetMap (OSM)
3. Collection Tools: Web scraping scripts (using BeautifulSoup/Scrapy) may be developed for automated data retrieval, complemented by manual downloads for verification.

3.5.2. Data Analysis Methods

1. Descriptive Statistics: Measures of central tendency and dispersion for facility characteristics.
2. Spatial Analysis: Network analysis for accessibility measurement, kernel density estimation for facility clustering, and choropleth mapping for visualization.

3. Equity Analysis: Calculation of Gini coefficients, Lorenz curves, and facility-to-population ratios.
4. Predictive Modeling: Linear regression for demand forecasting based on population trends.
5. Optimization Algorithms: Location-allocation modeling using p-median or maximal covering location problem formulations.

3.5.3. Tools for Analysis

- Data Processing: Python (Pandas, NumPy)
- Spatial Analysis: GeoPandas, PySAL, NetworkX
- Visualization: Matplotlib, Seaborn, Plotly, Folium
- Optimization: SciPy, PuLP
- Dashboard Development: Streamlit

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4. APPENDICES

4.1. APPENDIX A: PROPOSED SCHEDULE

4.2. APPENDIX B: TOOL(S) FOR DATA COLLECTION

4.3. APPENDIX C: PROPOSED BUDGET

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